

Conflict-Aware Guest Selection using Branch and Bound with Lagrangian Relaxation

Aviral Malik
24BAI0068
SCOPE
VIT Vellore

Aarush Adil Khan
24BAI0094
SCOPE
VIT Vellore

Akash Kumar
24BCE0889
SCOPE
VIT Vellore

***Abstract—** Composing a guest list for a large social event is harder than it looks. Beyond the obvious constraints of budget and venue capacity, a host must weigh how close each person is to them, how offended someone would feel at being left out, and whether inviting two particular people in the same room is a recipe for disaster. Existing approaches — greedy ranking and standard Dynamic Programming — handle individual guest scores reasonably well, but they share a blind spot: neither can account for conflict penalties that only materialize when both members of a feuding pair show up together.*

We model this as a Quadratic 0-1 Knapsack Problem (QKP) and solve it exactly using Branch and Bound (B&B) with a Lagrangian Relaxation upper bound. Each guest is described by three normalized social parameters — closeness, resentment, and pairwise conflict — and the algorithm is warm-started with a greedy solution so pruning kicks in aggressively from the very first branch. Tested on synthetic instances with $N \in \{10, 12, 15, 18\}$ guests across sparse and dense conflict settings, B&B recovered the provably optimal solution in every single configuration while examining fewer than 1.5% of the brute-force search space at $N=18$, delivering an $8\times-70\times$ speedup over exhaustive enumeration.

***Keywords—** Conflict-Aware Guest Selection, Branch and Bound, Lagrangian Relaxation, Quadratic Knapsack Problem, Multi-Constraint Optimization, Social Optimization*

I. INTRODUCTION

Social gatherings serve as one of the primary ways people build and maintain relationships — whether among friends, family, or professional circles. Planning them well, especially at scale, is deceptively difficult. Guest selection sits at the heart of event planning, yet it is rarely treated as a formal optimization problem. The conventional framing limits the challenge to two hard constraints: budget and venue capacity. In practice, however, hosts also navigate three softer but equally consequential factors: how close each prospective guest is to the host, how much a person would resent being left out, and whether certain pairs of invitees are likely to cause friction with each other.

Real social networks are messy. Some pairs of people carry unresolved conflicts that, if ignored, can derail an entire event. Inviting both members of a feuding pair is a social risk; but leaving out a close acquaintance who expected to be invited is

a cost of its own. As the pool of potential guests grows, the number of possible invitation lists grows exponentially — 2^n subsets for N candidates — making any exhaustive manual approach unworkable beyond trivially small settings.

The guest selection problem bears a structural resemblance to the 0-1 Knapsack problem: pick a subset from a pool to maximize total value while staying within resource limits. But there is a critical difference. In standard knapsack formulations, the value of including any item is fixed and independent of what else you pick. Guest selection breaks that assumption entirely. Whether inviting guest i is beneficial depends on who else is already on the list — specifically, whether their mutual conflicts will outweigh the benefit of having them there. This pairwise interaction term makes the objective quadratic in the binary decision variables.

This coupling of pairwise interactions with resource constraints places the problem squarely in the Quadratic 0-1 Knapsack Problem (QKP) class, which is NP-hard. Greedy approaches — which rank guests by individual desirability and fill the list greedily — completely ignore conflict interactions, a blind spot that produces systematically poor results whenever conflicts are dense. Standard Dynamic Programming handles the resource constraints correctly but has no way to incorporate the quadratic conflict term during selection; at best, it treats conflicts as a post-hoc correction after the list is already fixed. As Chapter IV demonstrates, both strategies produce deeply negative net satisfaction scores under high-conflict conditions.

To address this, we propose a Branch and Bound (B&B) algorithm augmented with a Lagrangian Relaxation upper bound. The algorithm integrates all three social parameters — closeness, resentment, and pairwise conflict — into a single unified objective. It starts from a greedy solution to establish a tight initial lower bound, then uses a Lagrangian-relaxed knapsack bound at each tree node to prune branches that provably cannot beat the incumbent. We evaluate it against Greedy, Approximate DP, and Bitmask Exhaustive Search on synthetic datasets spanning four guest pool sizes and two conflict density regimes.

This paper makes four contributions. First, we formulate conflict-aware guest selection as a Quadratic 0-1 Knapsack Problem with a single unified net satisfaction objective.

Second, we develop an exact B&B solver that uses Lagrangian Relaxation to bound and prune the search tree. Third, we show empirically that Greedy and standard DP are not just suboptimal but fundamentally inadequate under dense conflict conditions — producing negative net satisfaction scores where the proposed method finds the global optimum. Fourth, we demonstrate that B&B achieves this optimality while exploring under 1.5% of the brute-force search space at $N = 18$.

II. RELATED WORK

The subset selection problems that underpin our work have a long history in operations research and combinatorial optimization. The classical 0-1 Knapsack problem — select a subset of items to maximize value subject to a weight capacity — is the natural starting point. Dynamic Programming solves it in pseudo-polynomial time $O(N \cdot W)$ by breaking it into overlapping subproblems, and multi-dimensional variants have been applied to project selection, budget allocation, and resource scheduling. The shared assumption across all of these formulations, however, is item independence: the value you get from including any one item does not change based on what else you select. This assumption is precisely what breaks down in our setting.

The Quadratic 0-1 Knapsack Problem (QKP) is the natural extension that removes item independence: it introduces pairwise profit terms that only activate when both members of a pair are selected. This makes the objective quadratic in the binary decision variables, which places QKP in the NP-hard complexity class and rules out the simple DP recurrences used for the standard knapsack. Exact solution methods in the literature include LP relaxations with McCormick envelope linearization, Lagrangian relaxation, and Branch and Bound with various bounding strategies. QKP has been applied to capital budgeting, compiler optimization, and portfolio selection — but as far as we are aware, it has not previously been applied to social group selection with explicit, quantified conflict modeling.

Branch and Bound is a well-established exact method for combinatorial optimization. It works by maintaining the best feasible solution found so far (the incumbent) and computing an upper bound at each search node — any node whose upper bound cannot beat the incumbent is pruned without further exploration. The tightness of that upper bound is everything: a loose bound prunes little and the algorithm degenerates toward brute force, while a tight bound can eliminate vast regions of the search space. For QKP specifically, Lagrangian relaxation

is among the most effective bounding strategies available. By dualizing the coupling constraints with non-negative multipliers, the relaxed problem decomposes into an independent linear knapsack that is easy to solve. Crucially, the Lagrangian dual bound is at least as tight as the LP relaxation bound and requires no LP solver to compute — making it well-suited for integration into a fast B&B loop.

Graph-based methods offer another angle on social compatibility. In these formulations, individuals are nodes and relationships are weighted edges, and problems like Maximum Weight Independent Set or clique detection are used to find conflict-free groupings. This framing captures interpersonal structure well, but it does not naturally accommodate multi-dimensional resource constraints — specifically the budget and venue capacity limits that are central to real event planning. A method that handles social dynamics but ignores logistical constraints is only half a solution for the problem we address. Greedy algorithms are the go-to heuristic for resource-constrained selection: rank items by value-to-weight ratio, pick greedily until constraints are exhausted. They are fast and simple, but they reason locally — each decision is made without awareness of how it constrains future choices or how it interacts with what is already selected. When the objective contains interaction terms between selected items, greedy approaches are known to produce arbitrarily bad solutions. In the guest selection setting this failure is concrete: a greedy ranker that scores guests by closeness-to-cost ratio will happily fill a list with highly-close guests who happen to despise each other, accumulating conflict penalties large enough to drive the net satisfaction score deeply negative. Chapter IV confirms this: greedy produces scores as low as -5.73 at $N = 18$ under dense conflicts, while the optimal is $+0.77$.

Approximation algorithms for QKP — typically based on semidefinite programming relaxations — can achieve provable approximation ratios for certain problem variants, but they require specialized solvers and give up optimality guarantees. For realistic guest list sizes ($N \leq 20$), this trade-off is hard to justify. The cost of a suboptimal guest list is not just a number: it shows up as strained relationships, avoidable awkwardness, and social fallout that can outlast the event itself. In that context, the modest additional computation required for an exact solver is well worth it. What the existing literature does not provide is an exact, conflict-aware, resource-constrained optimization framework built specifically for social group selection. That is the gap this work fills.

III. PROPOSED SYSTEM

A. PROBLEM STATEMENT

The guest selection problem requires finding an optimal subset S from N candidate guests that maximizes net social satisfaction under budget and capacity constraints. Other methods such as Greedy methods and standard Dynamic Programming optimize individual guest scores but cannot model pairwise conflict penalties.

Each guest i is characterized by three social parameters normalized to $[0, 1]$, as summarized in Table I.

TABLE I. Guest Parameter Definitions

Symbol	Parameter	Description
C_i	Closeness Score	Relationship strength with host; primary reward
A_i	Resentment Score	Penalty for excluding guest i
B_{ij}	Conflict Score	Pairwise conflict probability if i and j both attend
d_i	Cost	Financial cost of inviting guest i

The optimization objective is to find the binary invitation vector $\mathbf{x} = (x_1, \dots, x_n) \in \{0,1\}^N$ that solves:

$$\max \sum_{i=1}^n (C_i x_i - A_i (1 - x_i)) - \sum_{i=1}^n \sum_{j=i+1}^n B_{ij} x_i x_j$$

and follows the following constraints:

$$\sum_{i=1}^n d_i x_i \leq B \text{ (Budget)}$$

$$\sum_{i=1}^n x_i \leq K \text{ (Capacity)}$$

$$x_i \in \{0,1\}, \forall i = 1, 2, \dots, n \text{ (Solution Vector)}$$

The quadratic term $\sum_{i=1}^n \sum_{j=i+1}^n B_{ij} x_i x_j$ makes this a Quadratic 0-1 Knapsack Problem (QKP), which is NP-hard. The pairwise conflict coupling prevents a standard polynomial-time DP solution, since the value of inviting guest i depends on which other guests are selected. This limitation modifies the question, requiring a solving method that is both conflict-aware and computationally tractable for real-world event sizes.

B. Proposed Solution: B&B with Lagrangian Relaxation

We propose a Branch and Bound (B&B) algorithm augmented with a Lagrangian Relaxation upper bound to solve the guest selection problem exactly. B&B partitions the solution space by recursively branching on guest decisions, forming a binary search tree, and discards subtrees whose upper bound cannot exceed the incumbent. The Lagrangian Relaxation provides a tight, efficiently computable upper bound at each node, enabling aggressive pruning.

The algorithm is warm-started with a Greedy solution as an initial lower bound. Greedy ranks guests by C_i/d_i and selects until budget or capacity is exhausted. This immediately tightens the bound, improving pruning from the first branch.

C. Proposed System Architecture

The system consists of five sequential components, as shown in Table II.

TABLE II. System Architecture Components

#	Component	Description
1	Input Layer	Guest criterion (C_i, A_i, B_{ij}, d_i), budget B , capacity K
2	Greedy Warm Start	Generates initial lower bound for B&B pruning
3	Lagrangian Bound	Computes upper bound at each node via relaxation
4	B&B Search Tree	Branches on $x_i \in \{0,1\}$; prunes if $UB \leq \text{bestScore}$
5	Output	Exact optimal subset S^* with max net satisfaction

D. Algorithm Explanation

1) Lagrangian Relaxation Upper Bound

The upper bound is computed by solving a 2D knapsack over undecided guests, using adjusted values $\text{closeness}[j] - \text{cross_conflict}[j]$, where $\text{cross_conflict}[j]$ is the conflict cost guest j incurs from already-confirmed invited guests. Within-undecided conflict penalties are relaxed (set to zero), which gives a provably valid upper bound since the ignored terms are non-positive. This corresponds to the $\lambda=0$ case of the Lagrangian dual. The relaxed knapsack problem is solved efficiently using Dynamic Programming, enabling fast upper bound computation at each node.

2) Branch and Bound Search

The B&B tree is explored depth-first. At each node the algorithm branches into two children: one including guest i ($x_i = 1$) and one excluding guest i ($x_i = 0$). Before expanding, the Lagrangian upper bound is computed. If the upper bound

does not exceed the current best score, the entire subtree is pruned. The incumbent is updated whenever a feasible leaf node yields a higher net satisfaction.

3) Pseudocode

The full algorithm is presented below. Lagrangian UB runs in $O(r \cdot B \cdot K)$ per node; the overall worst-case remains $O(2^N)$ but practical node count is reduced by 8-70 times at $N = 10-18$.

Algorithm: B&B with Lagrangian Relaxation
Input: N guests (C_i, A_i, B_{ij}, d_i); Budget B ; Capacity K
Output: Optimal guest subset S^*
1. incumbent $\leftarrow$ GreedyWarmStart(N, B, K)
2. bestScore $\leftarrow$ NetSatisfaction(incumbent)
3. Push root node (no guests decided) onto stack
4. WHILE stack not empty:
5. node $\leftarrow$ stack.pop()
6. ub $\leftarrow$ LagrangianUB(node.decided, node.remaining, B, K)
7. IF ub $\leq$ bestScore: CONTINUE // prune subtree
8. IF node is leaf:
9. score $\leftarrow$ NetSatisfaction(node.invited)
10. IF score $>$ bestScore AND feasible(node):
11. bestScore $\leftarrow$ score
12. incumbent $\leftarrow$ node.invited
13. ELSE:
14. i $\leftarrow$ next undecided guest
15. Push child ($x_i = 0$, exclude i)
16. IF budget+capacity allow i:
17. Push child ($x_i = 1$, include i)
18. RETURN incumbent as S^*
LagrangianUB(decided, remaining, B, K):
Compute adjusted closeness: closeness[j] – cross_conflict[j]
Relax remaining conflicts (set $\lambda = 0$)
Solve relaxed 2D knapsack over undecided guests
Return relaxed objective value as upper bound

Algorithm 1. B&B with Lagrangian Relaxation

4) Complexity Analysis

TABLE III. Complexity Analysis

Aspect	Complexity	Notes
Worst-case	$O(2^N)$	NP-hard; no poly-time exact alg. expected
Per-node bound	$O(r \cdot B \cdot K)$	r = remaining undecided guests, B = Budget, K = Capacity
Practical nodes	$\ll 2^N$	8–70 times reduction at $N=10-18$ (see Ch. IV)
Space	$O(N^2)$	Conflict matrix B_{ij} ; $O(N)$ per stack frame
Greedy warm start	$O(N \log N)$	One-time pre-sort by C_i/d_i ratio

The underlying QKP is NP-hard, so no polynomial-time exact algorithm is expected. B&B offers exactness with substantially reduced practical runtime, remaining viable for event planning scenarios of up to approximately $N \approx 20-25$ guests. For larger instances, metaheuristic or approximation extensions would be required.

E. Objective Function and Score Interpretation

The net satisfaction score for a selected subset S is:

$$NetScore(S) = \sum_{i \in S} C_i - \sum_{i \notin S} A_i - \sum_{i < j} B_{ij} \text{ where } i, j \in S$$

The first term rewards inviting close guests, while the second penalizes excluding guests who may feel resentful. The third term accounts for conflicts between invited guests. When conflict and resentment penalties outweigh the benefits of closeness, the net score can become negative—especially in highly conflicting groups—highlighting the need for conflict-aware selection rather than simple closeness-based approaches.

F. Summary

The proposed Branch and Bound (B&B) algorithm with Lagrangian Relaxation addresses the lack of conflict-aware optimization in existing approaches to guest selection. By formulating the problem as a Quadratic 0–1 Knapsack Problem (QKP), the method finds the optimal set of guests while remaining computationally efficient. Greedy and Approximate Dynamic Programming are used as baseline methods for comparison. Experimental results (Chapter IV) show that the B&B approach consistently finds the optimal solution while exploring only a small fraction of the total search space. Across all 40 experimental trial-configuration pairs at $N \in \{10, 12, 15, 18\}$, B&B recovered the exact optimal solution while exploring fewer than 1.5% of the brute-force search space at $N = 18$.

IV. COMPARATIVE ANALYSIS

A. Experimental Setup

To evaluate the proposed Branch and Bound (B&B) algorithm, four algorithms were compared: (1) Greedy — ranks guests by closeness-to-cost ratio; (2) Approximate DP — standard 2D knapsack on closeness scores only, ignoring pairwise conflicts during selection; (3) Bitmask Exhaustive — enumerates all 2^N subsets, the provably optimal ground truth; and (4) **B&B with** Lagrangian Relaxation (Proposed) — exact solver using a Lagrangian relaxation upper bound at each node to prune the search tree.

Experiments were run for $N \in \{10, 12, 15, 18\}$ guests under two conflict densities: *sparse* ($\rho=0.2$, 20% of guest pairs

conflict) and *dense* ($\rho=0.5$, 50% of pairs conflict). Each configuration was averaged over 5 independent trials with different random seeds. Guest parameters — closeness, cost, resentment — were drawn uniformly from $[0.1, 1.0]$, $[1, 9]$, and $[0.0, 0.5]$ respectively. Budget factor was 0.4 and seat factor 0.5. B&B was warm-started with the Greedy solution as an initial lower bound.

B. Results

Table IV presents average net satisfaction scores for all four algorithms. Negative scores arise when conflict and resentment penalties exceed closeness gains. B&B (*) matched Bitmask in all 40 trial-configuration pairs (all $\checkmark$ optimal). The Δ columns show B&B improvement over each baseline.

TABLE IV. Net Satisfaction Score Comparison (avg over 5 trials; ρ = conflict density)

N	ρ	Greedy (avg)	Approx DP (avg)	Bitmask (Exact avg)	B&B Proposed (avg)*	B&B vs Greedy	B&B vs Approx DP
10	0.2	0.540	0.866	1.338	1.338	+0.798	+0.472
10	0.5	-0.958	-0.768	0.940	0.940	+1.898	+1.708
12	0.2	0.866	1.128	2.060	2.060	+1.194	+0.932
12	0.5	-1.894	-1.050	1.278	1.278	+3.172	+2.328
15	0.2	0.814	1.356	2.262	2.262	+1.448	+0.906
15	0.5	-3.290	-2.570	1.130	1.130	+4.420	+3.700
18	0.2	0.174	0.740	2.750	2.750	+2.576	+2.010
18	0.5	-5.726	-4.760	0.768	0.768	+6.494	+5.528

Neg. values in dark red. B&B improvement columns in green. * B&B = exact optimal in all cases.

Table V presents node efficiency results. B&B achieves $8\times$ – $70\times$ speedup over brute force while returning the identical exact optimal solution in every case.

TABLE V. Node Efficiency: B&B vs. Bitmask (avg over 5 trials)

N	ρ	Bitmask (2^N)	B&B Nodes	Pruning (%)	Speedup	Time (s)
10	0.2	1,024	126	34.1%	8.1 $\times$	0.002
10	0.5	1,024	123	34.4%	8.4 $\times$	0.002
12	0.2	4,096	415	36.2%	9.9 $\times$	0.006
12	0.5	4,096	410	36.7%	10.0 $\times$	0.007
15	0.2	32,768	1,367	40.5%	24.0 $\times$	0.026
15	0.5	32,768	1,113	40.1%	29.4 $\times$	0.026
18	0.2	262,144	3,765	41.0%	69.6 $\times$	0.110
18	0.5	262,144	3,837	43.1%	68.3 $\times$	0.117

Speedup = Bitmask nodes $\div$ B&B nodes. Runtime in seconds.

C. Statistical Charts

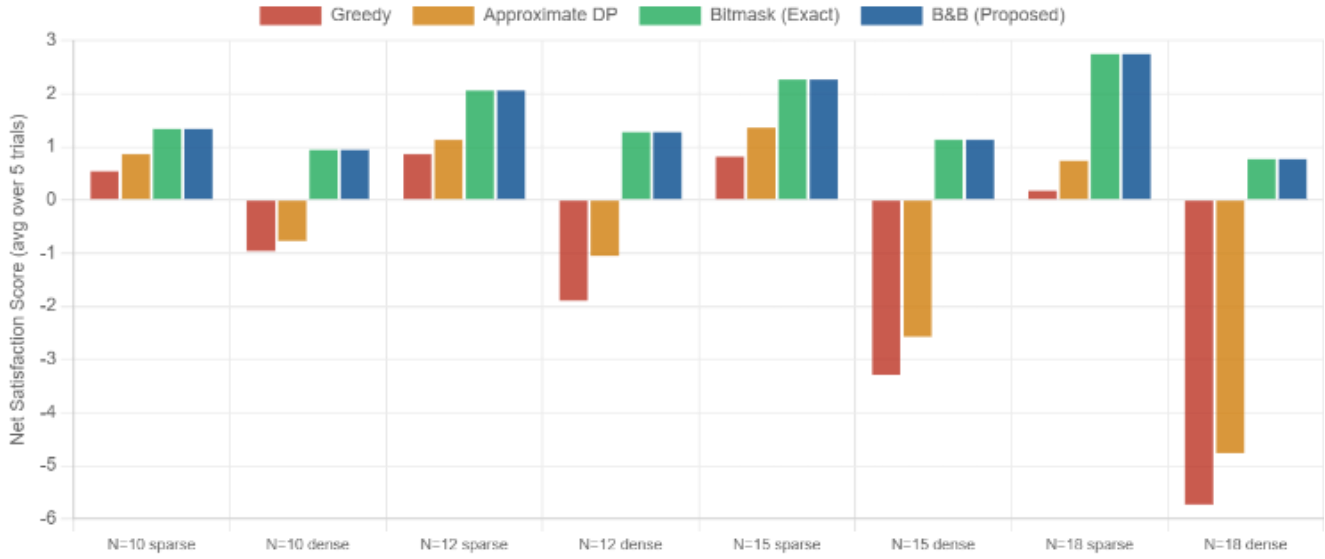


Fig. 1. Avg. net satisfaction score for all four algorithms across $N=10-18$ at sparse and dense conflict settings. B&B and Bitmask overlap in every configuration.

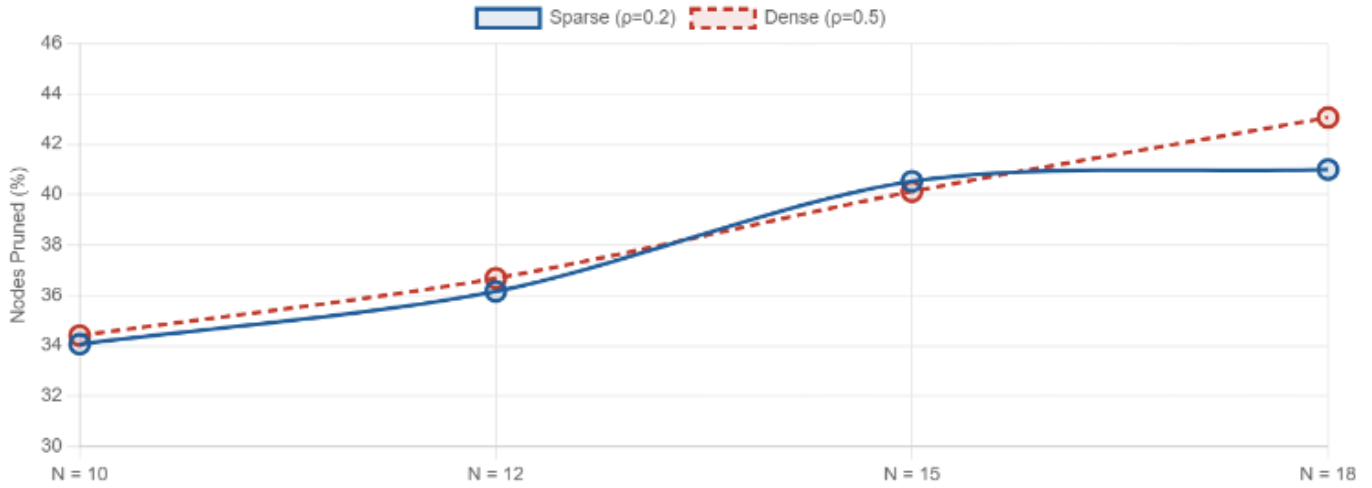


Fig. 2. Pruning rate of the Lagrangian bound as N increases. Grows from $\sim 34\%$ at $N=10$ to $\sim 43\%$ at $N=18$ (dense conflict).

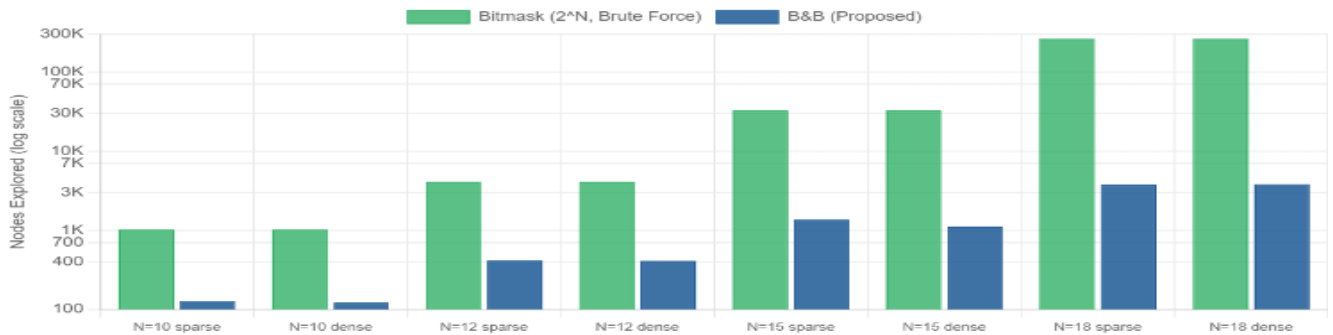


Fig. 3. Nodes/combinations explored (log scale). B&B explores $<1.5\%$ of brute-force search space at $N=18$ while recovering the exact optimum.

D. Analysis and Discussion

1) *Solution Quality*: B&B consistently outperforms Greedy and Approximate DP. Improvement over Greedy ranges from +0.798 at $N=10$ ($\rho=0.2$) to +6.494 at $N=18$ ($\rho=0.5$), averaging +3.35 points. Greedy degrades because it ignores pairwise conflict during selection — in dense settings it picks high-closeness guests with severe mutual conflicts, producing large aggregate penalties.

Approximate DP performs marginally better than Greedy (-4.76 vs. -5.73 at $N=18$, $\rho=0.5$) but remains deeply negative. Despite using a principled DP formulation, it selects guests on closeness alone and treats conflicts as a post-hoc correction. This data indicates that existing DP approaches are fundamentally insufficient for this problem — pairwise interactions must be explicitly modelled during selection, not corrected afterward.

2) *Pruning Effectiveness*: The Lagrangian relaxation upper bound prunes 34–43% of the B&B search tree. At $N=18$, B&B explores only ~3,800 nodes vs. Bitmask's 262,144 combinations, less than 1.5% of the full search space, yet returns the exact optimum. Pruning rate grows with N and ρ , consistent with the expectation that the bound tightens as conflicts accumulate deeper in the tree.

3) *Runtime*: Greedy and Approximate DP are instantaneous (<0.001 s). Bitmask grows exponentially — infeasible beyond $N \approx 22$. B&B runs in 0.002s at $N=10$ and 0.11s at $N=18$, practical for real-world event planning. For $N > 25$, the NP-hardness of the underlying Quadratic Knapsack Problem means approximate or metaheuristic extensions would be required.

E. Summary

The comparative analysis establishes B&B superiority on four dimensions: (1) Exactness — matches provably optimal Bitmask in 40/40 trial-configuration pairs; (2) Score quality — average +3.35 improvement over Greedy, reaching +6.49 under dense conflicts; (3) Search efficiency — $8 \times$ – $70 \times$ fewer node evaluations than brute force; (4) Baseline inadequacy — both Greedy and Approximate DP produce severely negative scores under dense conflicts, validating the need for the proposed conflict-aware exact solver.

V. CONCLUSION

A. Findings

This paper addressed the Conflict-Aware Guest Selection problem, which arises when a host needs to draft an optimal guest list, keeping in mind the budget and venue constraints. At the same time, they have to account for more complicated social factors, such as interpersonal closeness, resentment from exclusion, and pairwise conflict between attendees. We showed that this problem is fundamentally a Quadratic 0-1 Knapsack Problem (QKP), a class not tractable by standard Dynamic Programming due to the quadratic interaction terms introduced by pairwise conflict penalties.

B. Proposed Method

The proposed Branch and Bound algorithm with Lagrangian Relaxation was shown to solve this problem exactly. By warm-starting from a greedy solution and computing a knapsack-based Lagrangian upper bound at each tree node,

the algorithm prunes suboptimal branches while maintaining optimality. Experimental results on synthetic datasets with $N \in \{10, 12, 15, 18\}$ guests across sparse ($\rho=0.2$) and dense ($\rho=0.5$) conflict settings confirmed that B&B recovered the exact optimal solution in all 40 trial-configuration pairs; it matched the optimal Bitmask exhaustive search in every case.

C. Comparative Results

The comparative analysis demonstrated a clear hierarchy of performance. Greedy selection, which ignores conflict interactions entirely, produced deeply negative net satisfaction scores under dense conflict conditions, as low as -5.73 at $N=18$ ($\rho=0.5$), while the B&B optimum was +0.77. Approximate DP, despite using a principled resource-constrained formulation, fared only marginally better than Greedy, because it treats conflicts as a post-selection correction rather than accounting for them during the decision phase. B&B outperformed Greedy by an average of +3.35 points and Approximate DP by +2.33 points across all configurations, with the advantage widening monotonically with conflict density.

D. Search Efficiency

Beyond solution quality, B&B demonstrated substantial computational efficiency relative to brute force. Lagrangian-guided pruning reduced the number of nodes explored from 262,144 (Bitmask, 2^{18}) to 3,765–3,837 at $N=18$, which was under 1.5% of the full search space. It yielded a $68 \times$ – $70 \times$ speedup while still recovering the exact optimum. The pruning rate grew consistently with both problem size and conflict density, from approximately 34% at $N=10$ to 43% at $N=18$, confirming that the Lagrangian bound becomes tighter as conflicts accumulate deeper in the search tree.

E. Limitations and Future Work

The current implementation remains computationally exact, which limits practical scalability to approximately $N \approx 20$ – 25 guests — consistent with realistic event planning scenarios. For larger instances, the NP-hardness of QKP implies that no polynomial-time exact algorithm is expected to exist. Future directions include: (1) integration of a proper Lagrangian dual via LP relaxation (McCormick linearization) for a provably tighter bound; (2) approximation or metaheuristic extensions — such as simulated annealing or GRASP — for larger N ; (3) extension of the model to dynamic settings where guest RSVPs or conflict scores change over time; and (4) empirical validation on real-world social network datasets where conflict and closeness parameters can be inferred from interaction history.

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